
Narrowing Without Converging: How Interactive LLM Agents Search Under Feedback

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Abstract

1 Interactive agents should use feedback to converge on correct answers, yet how
2 they search remains poorly understood. We study three Qwen model tiers on 22
3 Turtle Soup puzzles using scored answer checkpoints and embedded trajectories
4 of successive questions. Accuracy improves for only the first 10 of 30 rounds,
5 while semantic stride contracts four- to fivefold. Longer budgets, Oracle quality,
6 and puzzle under-determination do not explain this plateau. Feedback also redi-
7 rects search poorly: after refutations, all three tiers move toward regions already
8 ruled out. Agents can also reach solution-grade stories and subsequently aban-
9 don them. These results identify two distinct failures, ineffective feedback use
10 and failure to preserve strong hypotheses, that endpoint accuracy cannot distin-
11 guish. We release the evaluation harness, process measures, verified puzzle set,
12 and per-game logs.

13 1 Introduction

14 Consider a Turtle Soup puzzle: *A man’s body lies in the desert, clutching half a matchstick; luggage*
15 *and clothing are scattered around him.* The solver sees only this surface and must recover the hidden
16 explanation through yes/no questions. In the solution, a hot-air balloon was losing altitude. After
17 discarding their luggage, the passengers drew lots, and the man holding the short matchstick was
18 thrown overboard.

19 Existing evaluations can determine whether an agent eventually recovers this story, but not how it
20 searches. The same endpoint score may describe an agent that never leaves its initial hypothesis
21 and one that discovers the answer but later abandons it. Existing Turtle Soup benchmarks empha-
22 size outcomes rather than search dynamics (Appendix A) [1, 2, 3]. This leaves a central question
23 unresolved: how does feedback change an agent’s next move?

24 Turtle Soup provides a controlled setting for answering this question. The interaction is textual, the
25 Oracle supplies a discrete yes/no/irrelevant signal, and each puzzle is deliberately under-determined.
26 We introduce two process-level probes. After every Oracle response, the Questioner produces a
27 complete checkpoint reconstruction, allowing accuracy to be tracked across rounds. We also embed
28 each question to trace the association trajectory and measure its per-round semantic stride. We
29 evaluate three Qwen model tiers on 22 human-played puzzles.

30 We test three hypotheses. **H1**: refutations progressively constrain the hypothesis space, so shorter
31 strides should accompany convergence. **H2**: feedback redirects the next question, with refutations
32 pushing the search away and confirmations promoting local refinement. **H3**: once an agent reaches
33 a solution-grade reconstruction, it preserves and commits that answer.

34 Our analysis shows that interactive search can narrow without converging. It also separates two fail-
35 ures hidden by endpoint accuracy: circling around an early hypothesis and abandoning a stronger

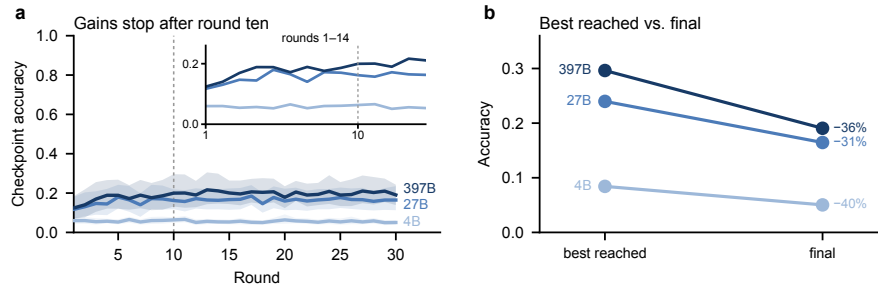


Figure 1: (a) Accuracy of the answer each tier would give at each round (bands, 95% cluster-bootstrap CIs over 22 puzzles); dashed rule at round 10; inset, rounds 1–14. (b) The best answer each game reached, against the answer it finally stands on.

earlier reconstruction. Finally, we establish an important measurement boundary: embedding geometry can describe these trajectories but cannot reliably score their success. We release the evaluation harness, process instruments, verified puzzle set, and per-game logs.

2 Framework and experiments

Task and harness. A **Questioner** sees only the puzzle’s visible *surface* and asks one yes/no question per round, following the SPLAT player–judge protocol [2]. An **Oracle** holds the hidden story and answers yes, no, or irrelevant. The Questioner may submit a final story at any round and must do so when the round budget is reached. The open-source harness records every exchange and the seed for each game.

Checkpoint and commitment instrumentation. In E1, the Questioner writes its best complete reconstruction after every Oracle response. We score each checkpoint, so every point on the accuracy curve represents a full answer based on the evidence available at that round. We separately record *commitment*, defined by whether and when the Questioner voluntarily submits a final story. Together, these measures convert an endpoint benchmark into a scored process.

Puzzles. To ensure that the intended explanations are demonstrably recoverable, we restrict evaluation to puzzles with a public record of human play. The resulting set contains 22 puzzles from a Chinese Turtle Soup community site, each retaining its source URL. Surfaces contain 9–108 characters, solutions contain 13–200 characters, and the set includes 97 annotated key clues. The repository separates this verified family from a quarantined generated family, and automated tests prevent the two from being mixed.

Puzzle under-determination. Each puzzle receives an **under-determination** score that measures how far the surface alone leaves a solver from the hidden solution. We generate twelve complete stories from the surface without Oracle feedback and identify the guess closest to the solution in embedding space. Under-determination equals one minus this maximum similarity, with higher values indicating a larger initial gap. We use the score as a puzzle-level covariate; Appendix B provides its distribution and validation.

Outcome scoring. The outcome score combines **clue recall** (70 points) and **causal-logic identity** (30 points), as detailed in Appendix C. The logic component recognizes causally equivalent answers expressed through different wording. The clue component anchors answers to the annotated evidence and prevents semantically distant alternatives from receiving high scores.

Association-trajectory instrumentation. Our second instrument analyzes the questions themselves, adapting forward flow [7] and the Divergent Association Task [8] from human psychometrics. For each round, we extract keywords with jieba, embed them using bge-small-zh-v1.5, and average the normalized embeddings into a unit vector q_t . All distances are cosine distances.

We use two anchors. The surface-only anchor represents terms from the visible puzzle surface, whereas the solution-aware anchor combines surface, solution, and clue terms. The primary measures are the **stride**, $s_t = d(q_t, q_{t-1})$, and the **anchor distance**, $h_t = d(q_t, \mathcal{A})$, computed separately for each anchor. H1 predicts that contraction in stride should accompany movement toward the solution and improved checkpoint accuracy. H2 is tested by pairing each Oracle response at round t with the next stride s_{t+1} and by comparing movement toward previously confirmed and refuted regions. H3 is tested by comparing solution-grade checkpoint peaks with subsequent commitment and final answers. The lexical `question_novelty` metric serves as an ablation that cannot capture movement expressed through synonyms.

Experimental designs. E1 plays each game for up to 30 rounds and scores a checkpoint after every round. E2 imposes hard caps of $\{5, 10, 15, 20, 25, 30\}$ rounds and scores only the forced final answer, providing a no-checkpoint control. E3 analyzes the E1 trajectories using $\{s_t\}$, $\{h_t\}$, response-conditioned movement, and checkpoint commitment. The keyword extractor is fixed across models, preventing any model from being evaluated through its own vocabulary.

Model grid and validity controls. We evaluate Qwen3.5-4B, Qwen3.6-27B, and Qwen3.5-397B-A17B on 22 puzzles with three seeds. E1 therefore contains 198 thirty-round games, while E2 contains 1,188 games across the six caps. DeepSeek-V3.1 serves as both Oracle and causal-logic judge. It is cross-family with every Questioner and achieves 95% accuracy on held-out yes/no probes. Every model call uses `max_tokens=2048`, disabled thinking, temperature 0.2, and a 50,000-token per-game budget. Analyses isolate the documented token-exhaustion artifact affecting the smallest tier; no harness failures occurred. Because the Questioners differ in training generation and architecture, we compare model tiers rather than claim a scaling law.

3 Results

Interaction improves accuracy only during the first ten rounds. Checkpoint accuracy increased through approximately round 10 and then plateaued at every tier (Figure 1a). From rounds 10 to 30, accuracy changed by -0.009 , $+0.003$, and -0.013 for 4B, 27B, and 397B, respectively; all confidence intervals included zero. Each tier also ended below its best checkpoint accuracy (Figure 1b).

The plateau was not explained by the round budget. Final accuracy was insensitive to the cap for the two larger tiers. The apparent decline at 4B coincided with token exhaustion, which rose from 0/66 to 63/66 games across caps; no decline appeared among non-exhausted games. The Oracle achieved 95% accuracy on held-out probes, and 68–74% of its responses were “no”, ruling out a collapsed feedback channel. The rank correlations between under-determination and round-1 accuracy were $-0.34/-0.54/-0.47$. No relationship was detected with improvement after round 10, any trajectory measure, either score component, or the between-tier gap (all n.s. at $n = 22$; Appendix C).

H1: the search narrows toward the answer. Mean stride decreased monotonically at every tier, falling four- to fivefold from 0.145/0.232/0.182 at round 1 to 0.025/0.053/0.045 at round 29. This contraction did not accompany convergence: accuracy remained flat, and within-tier stride did not predict final performance ($|\rho| \leq 0.16$). The 4B tier had the smallest late-game stride, 0.030 [0.021, 0.039], compared with 0.082 [0.065, 0.098] for 27B, yet never reached a solution-grade answer. Its trajectory therefore contracted around a persistent hypothesis rather than toward the solution. Excluding its 17% truncated or non-interrogative questions changed the mean and late stride only from 0.049 and 0.030 to 0.056 and 0.033, respectively.

H2: the answer redirects the next question. We paired each Oracle response with the next stride and controlled for round position within puzzle and round-bin cells. Let Δs denote stride after confirmation minus stride after refutation. No difference was detected for 4B ($+0.001$ [$-0.009, +0.011$]), whereas 27B and 397B moved farther after confirmation ($+0.028$ [$+0.010, +0.046$] and $+0.009$ [$+0.002, +0.017$]; Figure 2a). Response-conditioned movement was therefore absent at the smallest tier and opposite to the hypothesized direction at the larger tiers.

Because an informative confirmation could reasonably produce a larger step, we also used a direction-based test that assumed no optimal stride. After a refutation, the next question moved

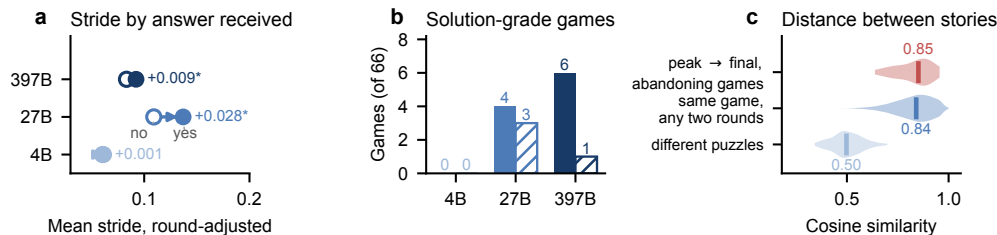


Figure 2: (a) Stride into the next question after refutation and confirmation, adjusted for round position. (b) Games reaching a solution-grade checkpoint and the subset voluntarily committed. (c) Similarity between final and peak stories in abandoning games relative to ordinary within-game variation.

toward regions the Oracle had already ruled out. This effect appeared at every tier: -0.089 $[-0.112, -0.068]$ for 4B, -0.051 $[-0.075, -0.032]$ for 27B, and -0.032 $[-0.047, -0.018]$ for 397B. No corresponding asymmetry appeared after confirmation. Feedback therefore failed to steer any tier consistently away from rejected regions.

H3: the agent stops when it finds the answer. We defined a sustained checkpoint peak of at least 0.5 as *solution-grade*. The 4B tier reached no solution-grade checkpoint. The 27B tier reached solution-grade checkpoints in four games and voluntarily committed in three; 397B did so in six games but committed in only one (Figure 2b). Commitment was rare and late overall, occurring in 11/66, 9/66, and 2/66 games at mean rounds 11.1, 14.2, and 23.5.

The dominant failure differed by tier. The 4B tier circled around an early hypothesis in 39/66 games and abandoned a stronger answer in one. The 397B tier circled in 6/66 games and abandoned a stronger answer in 12. The 397B tier therefore reached useful hypotheses more often but did not reliably preserve or commit them.

Embedding geometry describes abandonment but does not score it. In abandoning games, the median cosine similarity between the final and peak stories was 0.851. This was comparable to the similarity between checkpoints from the same game (0.841) and much greater than that between stories from different puzzles (0.498; Figure 2c). Despite this semantic proximity, abandoning the peak reduced the score by 0.30 on average. The decisive substitution was therefore smaller than the encoder’s resolution.

Consistent with this limit, no systematic drift was detected, and all within-tier coefficients linking stride or drift to best accuracy were bounded within ± 0.035 per standard deviation (Appendix C). Lexical novelty detected only the 4B tier’s repeated questions. The trajectory geometry characterized how agents failed, but it did not predict how well they scored.

4 Limitations and Conclusion

Limitations. Our evidence comes from one game, one Oracle protocol, and three Qwen-family Questioners. The trajectory measures capture how far each question moves in embedding space, not what operation it performs; round position is controlled only by binning, the anchors are not human-derived, and a single encoder may miss small semantic changes that alter correctness. Token exhaustion can mimic inability to form questions; the Oracle’s 95% audited accuracy implies roughly one wrong response per game; and using the same DeepSeek model as Oracle and logic judge may correlate their errors. Under-determination may partly measure puzzle familiarity, though no claim rests on it. The puzzles include death and other dark themes and carry content flags.

Conclusion. Here, the agents narrow without converging: semantic stride decreases four- to fivefold while accuracy plateaus, and feedback fails to steer questions consistently away from rejected regions. The smallest tier mostly circles early hypotheses, whereas the largest tier sometimes reaches solution-grade reconstructions but fails to preserve or commit them. The two failures require different fixes yet appear identical under endpoint accuracy; process-level measurement reveals how search fails even when it cannot predict performance. Whether these patterns extend to other tasks, model families, and protocols remains open.

Declaration of LLM usage

LLMs are the object of study: the Questioner agents, the Oracle, and the causal-logic judge are all LLMs, specified in §3 and Appendix C, and every reported number derives from their logged interactions. Separately, LLM assistants were used to help write analysis code and edit prose; the authors reviewed all analyses, claims, and text, and take full responsibility for them.

References

- [1] Q. Yu *et al.* TurtleBench: Evaluating Top Language Models via Real-World Yes/No Puzzles. arXiv:2410.05262.
- [2] Q. Chen, B. Zhang, G. Wang, Q. Wu. Weak-eval-Strong: Evaluating and Eliciting Lateral Thinking of LLMs with Situation Puzzles. arXiv:2410.06733.
- [3] M. Zhou *et al.* What to Ask Next? Probing the Imaginative Reasoning of LLMs with TurtleSoup Puzzles. arXiv:2508.10358.
- [4] J. Park, I. Baek, J. Park, H. Lee. Beyond One Path: Evaluating and Enhancing Divergent Thinking in Interactive LLM Agents. arXiv:2605.28465.
- [5] J. Ye *et al.* Justice or Prejudice? Quantifying Biases in LLM-as-a-Judge. ICLR 2025. arXiv:2410.02736.
- [6] L. Shi *et al.* Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge. IJCNLP-AACL 2025.
- [7] K. Gray *et al.* “Forward Flow”: A New Measure to Quantify Free Thought and Predict Creativity. *American Psychologist* 74(5), 2019.
- [8] J. A. Olson *et al.* Naming Unrelated Words Predicts Creativity. *PNAS* 118(25), 2021.
- [9] S. De Deyne *et al.* The “Small World of Words” English Word Association Norms. *Behavior Research Methods* 51(3), 2019.

182 A Related work

183 Existing turtle-soup benchmarks, TurtleBench [1], SPLAT [2] (whose interactive player-judge pro-
 184 tocol we adopt), and TurtleSoup-Bench [3], score verification accuracy, solve efficiency, or faithful-
 185 ness to the intended story. TurtleSoup-Bench probes what agents ask next but grades it against the
 186 intended story; none of the three measures the search’s geometry, which is our delta. MUTATE [4]
 187 shows frontier agents find far fewer mechanism-distinct paths than humans; its diverge-then-narrow
 188 agent is a natural test intervention for our metrics. The instrument itself adapts forward flow [7]
 189 and the Divergent Association Task [8] from human psychometrics. The Small World of Words
 190 norms [9] provide the human anchor that ours is not yet (§2). Judge-bias catalogues [5, 6] shape
 191 the scoring: the model judge is confined to the one subscore that objective matching cannot reach
 192 (Appendix C).

193 B The puzzle set

194 **Where the puzzles come from.** A puzzle nobody has solved carries no evidence that its clues
 195 suffice, that its solution is unique, or that the intended leap is reachable; an agent scoring zero on
 196 it teaches us nothing. Our set is hand-picked from the classic repertoire of a Chinese Turtle Soup
 197 community: 22 puzzles people have played and solved, each keeping its source record. Every item
 198 was read and kept or rejected by hand; rejected were puzzles turning on deduction from stated
 199 evidence rather than a lateral reframing, and puzzles whose surfaces were incomplete at the source.
 200 Surfaces run 9–108 characters (median 34), solutions 13–200 (median 102).

201 **How hard are they?** Each puzzle carries an **under-determination** score: give a model the surface
 202 alone, with no Oracle and no feedback, let it write twelve complete stories, and take the closest to
 203 the real solution. The index is one minus that best guess, and is large when the surface leaves a
 204 solver far from the answer.

Under-determination	Puzzles
0.15–0.25	5
0.25–0.35	8
0.35–0.45	4
0.45–0.55	4
0.55–0.65	1

205
 206 The median is 0.347, quartiles 0.273 and 0.410, full range 0.173–0.577: single-peaked with a thin
 207 hard tail. Easiest are the story the genre is named for (0.17) and the desert matchstick (0.19). Hardest
 208 is a fourteen-character surface reading *our heights differ / a flowerpot broke / our heights are the*
 209 *same* (0.58).

210 Two properties make this usable as a covariate. It describes the puzzle, not any agent’s performance
 211 on it. It is also independent of rounds used; a difficulty defined as “how many rounds this takes”
 212 would make an accuracy-versus-rounds plot confirm itself. Nor is it a proxy for surface length
 213 ($r = -0.22$, n.s.), so length dependence was not detected at $n = 22$ (a weak test). It is one measure
 214 rather than two: a surface loose enough to admit many mechanisms is, for that same reason, one
 215 whose true mechanism takes more reframing to reach, so *how far* and *how many ways* are not
 216 separable here.

217 **A caveat we report either way.** The two puzzles scoring easiest are among the most widely circu-
 218 lated, and the models’ own first-round guesses are consistent with familiarity contributing: on the
 219 best-known of them the two larger tiers open at 0.61 and 0.53 before any evidence arrives, against
 220 0.24 and 0.36 on a puzzle of almost identical under-determination. The measure may therefore read
 221 familiarity as much as difficulty on the easy end. Separating the two calls for a memorisation probe,
 222 which we have not run; nothing in this paper rests on the measure.

223 C Measuring agent performance

224 **What counts as a correct answer.** Judged outcome is **clue recall (70)** plus **causal-logic identity**
 225 **(30)**. Clue recall matches the answer against per-puzzle annotated key clues: objective and repro-
 226 ducible. The logic judge rates only whether cause $\rightarrow$ mechanism $\rightarrow$ outcome matches and is told

explicitly to ignore wording. Because single ratings are unstable on borderline answers, ratings are sampled and averaged (two samples per rating in the reported grid). The gap between the two subscores is itself readable: right story, wrong vocabulary.

Clue matching tolerates paraphrase through character-bigram recall, which sets a floor on how short an annotated clue may be. Below that floor only a verbatim match counts, and the score reads vocabulary rather than content. Clues are annotated above the floor.

Does the logic half earn its thirty points? On the full grid, yes: every model earns proportionally far more of the judged causal-logic half than of the clue-recall half (397B averages 12.6/30 logic against 7.2/70 clues, 27B 10.1 against 6.8, 4B 2.9 against 0.2), and the halves correlate only weakly per game (Spearman $\rho = 0.30/0.35/0.06$ for 397B / 27B / 4B over 396 E2 games each): models reach the right causal shape well before, and often without, landing the annotated clue content. The 70/30 weighting itself is a design choice; every conclusion in §3 is readable from the two halves separately, though the absolute plateau level (~ 0.2) does depend on the weights. The logic judge rates the causal-logic half on its 30-point scale and is sampled twice per rating at temperature 0.2; the two samples agree exactly in 86% of ratings and differ by under one point of 30 on average.

Feedback response: definition, confound, and both estimates. For each round t the Oracle’s answer is paired with s_{t+1} , the stride into the next question; “irrelevant” rounds are excluded. Confirmation arrives earlier than refutation (mean round 12.1 / 9.7 / 10.9 for a “yes” against 15.6 / 15.6 / 15.9 for a “no”) and early strides are larger, so an uncontrolled comparison is confounded by position. Pairs are formed inside (puzzle, round-bin) cells with bins {1–5, 6–10, 11–15, 16–20, 21–25, 26–29}, and 95% CIs bootstrap over puzzles (4,000 resamples). Uncontrolled, all three tiers appear to respond (+0.024 [+0.004, +0.044]; +0.043 [+0.015, +0.070]; +0.034 [+0.023, +0.046]); controlled, the smallest does not (+0.001 [−0.009, +0.011]; +0.028 [+0.010, +0.046]; +0.009 [+0.002, +0.017]). The gap between the two is the point.

The semantic scale of abandoning. Within abandoning games (sustained peak ≥ 0.3 , final below half of it; $n = 34$), the final checkpoint story sits at cosine 0.851 (median) from the peak story, while any two checkpoints of the same game sit at 0.841 (median over all 77,906 within-game pairs from 60 games) and two stories from different puzzles at 0.498 (231 pairs). The score gap across that same pair is $0.447 \rightarrow 0.099$.

The round-1 prior. The round-1 checkpoint precedes the first Oracle answer and is therefore a cold guess. It holds 49% / 33% / 26% (median, by tier) of the game’s eventual best score, and correlates with it at $\rho = +0.37/ +0.57/ +0.47$ (permutation $p = 0.09/0.003/0.03$, $n = 22$ puzzles with seeds averaged). Under-determination, itself measured from twelve cold guesses by a model cross-family with every Questioner, predicts round-1 quality at $\rho = -0.34/ -0.54/ -0.47$ ($p = 0.12/0.010/0.029$) but not the final score.

Behavioural indicators against puzzle structure. Regressing each per-trace indicator on under-determination, seeds averaged, 5,000 permutations: stride, late stride, drift (both anchors), mean anchor distance and rounds played are all n.s. at every tier ($|\rho| \leq 0.32$, $p > 0.10$). Commit rate is negative at all three (−0.32/ −0.25/ −0.43) but pooled $p = 0.13$, and 1 of 24 tests reaching $p < 0.05$ is the chance expectation, so we report it as unresolved rather than as an effect.

Retention estimator and permutation null. Retention is the mean per-game final-to-sustained-peak ratio over games with a nonzero sustained peak; the final is the last played checkpoint, and excluding early-committing games changes each tier’s value by ≤ 0.03 (bootstrap 95% CIs $\pm \approx 0.08$). For the null: unit, the game. For each of 2,000 permutations, every game’s checkpoint sequence is shuffled within the game (knowledge-constant, jitter-only null), the sustained peak and the final (last element) are recomputed on the shuffled sequence, and retention is the mean over that tier’s games with a nonzero permuted peak. The reported p is one-sided per tier, the fraction of permutations whose null retention is at or below the observed value. Observed retention (0.69–0.76 by tier) does not fall significantly below this null (0.79–0.84, $p \geq 0.09$), so the peak-to-final gap visible in Figure 1b is reported as consistent with scoring jitter and no claim in the body rests on its size; the commitment record of §3 is behavioural and unaffected by jitter. Late-game stride means the second half of a game’s strides.

Mixed-model coefficients. Best checkpoint accuracy per game; Gaussian LMM, puzzle random intercept, tier fixed effect, ML fits, likelihood-ratio tests against the covariate-only model. Stride

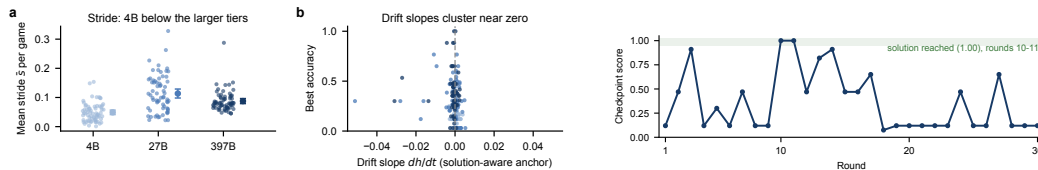


Figure 3: Per-trace geometry ($n = 198$). (a) Per-game mean stride by tier (dots; bars, tier mean $\pm$ 95% CI): 4B sits below the larger tiers. (b) Drift slope against best accuracy, clustered near zero at every tier.

Figure 4: One abandoning game (refsoup_021, 397B, seed 2): the round 10–11 checkpoint stories are the solution (band), never volunteered, abandoned by round 19. Full rows in Appendix C.

281 $\beta = -0.010$ per SD, 95% CI $[-0.035, 0.014]$, $p = 0.42$; drift $\beta = -0.011$, $[-0.035, 0.014]$,
 282 $p = 0.39$; stride $\times$ tier $p = 0.20$; surface-anchor drift $\beta = -0.020$, $[-0.044, 0.003]$, $p = 0.097$.

283 **Environment settings and Oracle audit.** Every model call uses max_tokens 2048 (thinking dis-
 284 abled, temperature 0.2) with a per-game budget of 50,000 tokens; exhausting the game budget forces
 285 a final answer and is recorded as the game outcome token_budget. The Oracle audit probes a
 286 candidate with 25 hand-crafted (question, expected-answer) pairs over three puzzles, 21 with a def-
 287 inite yes/no answer and 4 expecting “irrelevant”; the pass bar is 90% on the yes/no items. The
 288 deployed DeepSeek-V3.1 scored 20/21 (95%) on yes/no and 3/4 exact on “irrelevant”. Candidates
 289 measured in the pilot phase: a 4B Oracle with thinking disabled, 30% (rejected; in that pilot config-
 290 uration, raising Oracle accuracy from 30% to 100% left the 4B Questioner’s score unchanged); the
 291 same model with thinking enabled, 70% (rejected); a same-family flash model, 90% (excluded for
 292 sharing the Questioners’ family); GLM-5.3-flash, 100% (used in earlier tooling).

293 **Trajectory and difficulty pipeline.** Keywords are extracted with jieba after stripping Latin char-
 294 acters and digits; each keyword is embedded with bge-small-zh-v1.5 (normalized), and the round
 295 vector is the re-normalized mean of its keyword embeddings. All distances are cosine. The solution-
 296 aware anchor embeds the puzzle’s surface, solution, and clue terms; the surface-only anchor embeds
 297 surface terms alone. The under-determination measure prompts GLM-5.3-flash (cross-family with
 298 all Questioners) for twelve cold guesses from the surface and scores closeness as embedding sim-
 299 ilarity between guess and solution under the same encoder, not with the composite score, so the
 300 covariate is not coupled to the outcome apparatus.

301 **Artifacts.** Harness, puzzle set with per-item source records, per-game logs (including every check-
 302 point story), and the analysis scripts behind each reported number are released at a public repository,
 303 withheld here for double-blind review and available to reviewers on request through the programme
 304 chairs. Puzzle surfaces and solutions originate from a public community repertoire and are redis-
 305 tributed with source attribution and a content flag.

306 **Where the accuracy curve comes from.** E1 does not score once at the end. Every round, after
 307 the Oracle answers, the Questioner is required to commit its best complete story, and *that* is scored.
 308 Each point on the accuracy curve is therefore a full answer written with the evidence available at
 309 that round.

310 **One game, worked.** Puzzle refsoup_021 (surface: a stormy night, a run-down temple, two
 311 photographs on the wall that seem to stare; by morning the narrator sees only two windows; solution:
 312 “there were never photos on the wall; people were pressing against the windows, looking in”), played
 313 by Qwen3.5-397B, seed 2. Table 1 shows seven of thirty rounds; Questions and stories are our
 314 translations of the logged Chinese, stories abridged to the clause carrying the hypothesis. Rounds
 315 22–30, eight lexically distinct re-asks of the same flooding hypothesis (e.g. round 25, whether the
 316 narrator woke floating upside down; round 28, whether rising water reached eye level), are in the
 317 released logs.

318 Three things the table shows that no mean curve can. **The checkpoint mechanism surfaces knowl-**
 319 **edge the agent never acts on.** The round-10 committed story *is* the solution, synthesized from
 320 earlier answers on a round whose own question was answered “no”. Yet the model never volun-
 321 teers a final answer in all thirty rounds (one of the 57/66 non-committing 397B games), and by
 322 round 19 it has abandoned the correct reading for a falling-coverings mechanism it never escapes.
 323 **Stalling in fresh words happens at the largest scale too.** Rounds 22–30 are eight lexically distinct

Rd	Question (abridged)	Ans.	Committed story (abridged)	score (clue+logic)
1	Are the ones in the photos people?	yes	The two “photos” are two people nailed or sealed into wall niches.	0.12 (0+12)
3	On waking the next day, is he still in the temple?	yes	The “two photos” are really two windows, mistaken for photos in the dim light.	0.91 (70+21)
4	Are the “two windows” the same objects as last night’s “photos”?	yes	The “photos” were hollowed-out window openings; the fear came from darkness and illusion.	0.12 (0+12)
10	Did frame glass break and fall, leaving open holes?	no	They were never photos, but faces pressed against the outside of the windows.	1.00 (70+30)
11	Were the photos removed, revealing windows behind?	no	The staring figures were people outside the windows.	1.00 (70+30)
19	Were real people hiding behind the photos?	no	Window openings sealed by painted cloth covers; the storm tore them off.	0.12 (0+12)
30	Did rising water lift the narrator higher?	no	Papered-over window openings; the storm stripped the paper away.	0.12 (0+12)

Table 1: Seven rounds of one E1 game (refsoup_021, Qwen3.5-397B, seed 2): the Questioner’s question, the Oracle’s answer, the story the agent was forced to commit that round, and its composite score. Questions and stories are our translations; the original transcripts are in the released logs.

re-asks of the same flooding hypothesis, exactly the pattern lexical novelty misses. **The curve’s round-to-round jitter is a scoring fact, not a knowledge fact.** The same understanding re-worded (rounds 3 $\rightarrow$ 4) swings the clue half 70 $\rightarrow$ 0. The mean curves of §3 average this out, which is why single-game curves are not the unit of analysis.